

ТЗ.

ЭТО КОСМОС

$$p(x) = \begin{cases} e^{-\frac{x}{\theta}}, & x \geq 0, \\ 0, & x < 0 \end{cases} \quad \theta > 0, \quad n = 3$$

$$\tilde{\theta}_1 = \bar{x}, \quad \tilde{\theta}_3 = x_{(2)}$$

$$\tilde{\theta}_1 = \bar{x} = \frac{1}{n} \sum_i x_i$$

а) проверить на несмещенность

$$\begin{aligned} 1) M\tilde{\theta}_1 &= \int_{-\infty}^{+\infty} x d f(x) = \int_0^{+\infty} \frac{x}{\theta} e^{-\frac{x}{\theta}} dx = -\frac{\theta}{\theta} \int_0^{+\infty} x d e^{-\frac{x}{\theta}} = \\ &= -x e^{-\frac{x}{\theta}} \Big|_0^{+\infty} + \int_0^{+\infty} e^{-\frac{x}{\theta}} dx = -\theta e^{-\frac{x}{\theta}} \Big|_0^{+\infty} = \theta \end{aligned}$$

$$M\tilde{\theta}_1 = M\left(\frac{1}{n} \sum_i x_i\right) = \frac{1}{n} \cdot n M\tilde{\theta}_1 = M\tilde{\theta}_1 = \theta$$

⇒ несмещенность

$$\begin{aligned} M\tilde{\theta}_1^2 &= \int_{-\infty}^{+\infty} x^2 d f(x) = \frac{1}{\theta} \int_0^{+\infty} x^2 e^{-\frac{x}{\theta}} dx = -\frac{\theta}{\theta} \int_0^{+\infty} x^2 d e^{-\frac{x}{\theta}} = \\ &= -e^{-\frac{x}{\theta}} \cdot x^2 \Big|_0^{+\infty} + 2 \int_0^{+\infty} e^{-\frac{x}{\theta}} x dx = 2\theta \int_0^{+\infty} x d e^{-\frac{x}{\theta}} = \\ &= -2\theta x e^{-\frac{x}{\theta}} \Big|_0^{+\infty} + 2\theta \int_0^{+\infty} e^{-\frac{x}{\theta}} dx = -2\theta^2 e^{-\frac{x}{\theta}} \Big|_0^{+\infty} = \\ &= 2\theta^2 \end{aligned}$$

$$D\tilde{\theta}_1 = M\tilde{\theta}_1^2 - M^2\tilde{\theta}_1 = 2\theta^2 - \theta^2 = \theta^2$$

$$2) \tilde{\theta}_2 = x_{(2)}$$

$$f(x) \sim \sum_{i=k}^n C_n^i f^i(t) (1-f(t))^{n-i}$$

$$\zeta(2) \sim C_3^2 f^2(x)(1-f(x)) + C_3^3 f^3(x) = 3f^2(x)(1-f(x)) + f^3(x) = 3f^2(x) - 2f^3(x) = \psi(x)$$

$$p(x) = \frac{e^{-\frac{x}{\theta}}}{\theta} \cdot f(x) = \int_0^x p(t) dt = \left[-e^{-\frac{t}{\theta}} \right]_0^x = (1 - e^{-\frac{x}{\theta}})$$

$$\psi'(x) = 6 f(x) \cdot p(x) - 6 f^2(x) \cdot f(x) = \psi(x) \Rightarrow$$

$$\Rightarrow \psi(x) = 6(1 - e^{-\frac{x}{\theta}}) \cdot \frac{e^{-\frac{x}{\theta}}}{\theta} \cdot (1 - 1 + e^{-\frac{x}{\theta}}) = \frac{6}{\theta} (e^{-\frac{2x}{\theta}} - e^{-\frac{3x}{\theta}})$$

$$M\tilde{\Theta}_2 = Mx_{(2)} = \int_{-\infty}^{+\infty} \frac{6x}{\theta} (e^{-\frac{2x}{\theta}} - e^{-\frac{3x}{\theta}}) dx =$$

$$= 6 \int_{-\infty}^{+\infty} x \cdot \frac{e^{-\frac{2x}{\theta}}}{\theta} dx - 6 \int_{-\infty}^{+\infty} x \cdot \frac{e^{-\frac{3x}{\theta}}}{\theta} dx =$$

$$= -\frac{6}{\theta} \cdot \frac{\theta}{2} \int_0^{+\infty} x de^{-\frac{2x}{\theta}} + \frac{6}{\theta} \cdot \frac{\theta}{3} \int_0^{+\infty} x de^{-\frac{3x}{\theta}}$$

$$= -3x e^{-\frac{2x}{\theta}} \Big|_0^{+\infty} + 3 \int_0^{+\infty} e^{-\frac{2x}{\theta}} dx + 2x e^{-\frac{3x}{\theta}} \Big|_0^{+\infty} +$$

$$+ 3 \int_0^{+\infty} e^{-\frac{2x}{\theta}} dx - 2 \int_0^{+\infty} e^{-\frac{3x}{\theta}} dx = \frac{3\theta}{2} - \frac{2\theta}{3} =$$

$$= \frac{5}{6}\theta \Rightarrow \text{оценке не}$$

$$\tilde{\Theta}_2' = \frac{6}{5} \tilde{\Theta}_2 \Rightarrow M\tilde{\Theta}_2' = \theta - \text{несмещен}$$

сравнение экстремальности.

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$$1) \tilde{\Theta}_1: D\tilde{\Theta}_1 = D\left(\frac{1}{n} \sum_i x_i\right) = \frac{D\beta}{n} = \frac{\Theta^2}{n} = \frac{\Theta^2}{3}$$

$$2) M\tilde{\Theta}_2^2 = 6 \int_0^{+\infty} x^2 \frac{e^{-\frac{2x}{\Theta}}}{\Theta} - 6 \int_0^{+\infty} x^2 \frac{e^{-\frac{3x}{\Theta}}}{\Theta} =$$

$$= -\frac{6 \cdot \Theta}{\Theta \cdot 2} \int_0^{+\infty} x^2 \cdot d e^{-\frac{2x}{\Theta}} + \frac{6 \cdot \Theta}{3 \cdot \Theta} \int_0^{+\infty} x^2 d e^{-\frac{3x}{\Theta}} =$$

$$= -3 \int_0^{+\infty} x^2 d e^{-\frac{2x}{\Theta}} + 2 \int_0^{+\infty} x^2 d e^{-\frac{3x}{\Theta}} =$$

$$= -3x^2 e^{-\frac{2x}{\Theta}} \Big|_0^{+\infty} + 6 \int_0^{+\infty} x e^{-\frac{2x}{\Theta}} dx + 2x^2 e^{-\frac{3x}{\Theta}} \Big|_0^{+\infty} -$$

$$- 4 \int_0^{+\infty} x e^{-\frac{3x}{\Theta}} dx = -\frac{6\Theta}{2} \int_0^{+\infty} x d e^{-\frac{2x}{\Theta}} + \frac{4\Theta}{3} \int_0^{+\infty} x d e^{-\frac{3x}{\Theta}} =$$

$$= -\frac{6\Theta}{2} x e^{-\frac{2x}{\Theta}} \Big|_0^{+\infty} + 3\Theta \int_0^{+\infty} e^{-\frac{2x}{\Theta}} dx + \frac{4}{3}\Theta x e^{-\frac{3x}{\Theta}} \Big|_0^{+\infty} -$$

$$- \frac{4}{3}\Theta \int_0^{+\infty} e^{-\frac{3x}{\Theta}} dx = -\frac{3\Theta^2}{2} e^{-\frac{2x}{\Theta}} \Big|_0^{+\infty} + \frac{4}{3}\Theta^2 e^{-\frac{3x}{\Theta}} \Big|_0^{+\infty} =$$

$$= \frac{19}{18} \Theta^2 = M\tilde{\Theta}_2^2 \Rightarrow$$

$$D\tilde{\Theta}_2 = M\tilde{\Theta}_2^2 - M^2\tilde{\Theta}_2 = \frac{19}{18} \Theta^2 - \frac{25}{36} \Theta^2 = \frac{13}{36} \Theta^2$$

$$D\tilde{\Theta}_1 = \frac{36}{25} D\tilde{\Theta}_2 = \frac{13}{25} \Theta^2$$

$$D\tilde{\Theta} = \frac{\Theta^2}{3}, \quad \forall \Theta \in \mathbb{R} \in (0; +\infty): D\tilde{\Theta}_1 < D\tilde{\Theta}_2$$

$$D\tilde{\Theta}_1 = \frac{13}{25} \Theta^2 \quad \tilde{\Theta}_1 - \text{экстремальнее } \tilde{\Theta}_2$$

• неравенство К-Ю.

проверки: 1) регулярность можно

2) регулярность огибающей

$$1) f(x, \theta), \theta \in \Theta \subset \mathbb{R}; f(x, \theta) = (1 - e^{-\frac{x}{\theta}}) \mathbb{I}_{(0, +\infty)}$$

- непрерывно зависит на $\theta \in \Theta \subset \mathbb{R}^3$

$$\frac{\partial}{\partial \theta} \int_{-\infty}^{+\infty} df(x, \theta) = \int_{-\infty}^{+\infty} \frac{\partial f(x, \theta)}{\partial \theta} dx$$

$$\int \frac{\partial p(x, \theta)}{\partial \theta} dx = 0$$

$$A(\theta) \int_0^{+\infty} \frac{\partial \left(\frac{1}{\theta} e^{-\frac{x}{\theta}} \right)}{\partial \theta} dx = \int_0^{+\infty} \frac{x \cdot e^{-\frac{x}{\theta}}}{\theta^3} - \int_0^{+\infty} \frac{e^{-\frac{x}{\theta}}}{\theta^2} dx =$$

$$= \frac{1}{\theta} - \frac{1}{\theta} = 0$$

интер. Фаннера: $I(\theta) = M \left[\left(\frac{\partial \ln p(x, \theta)}{\partial \theta} \right)^2 \right] =$

$$= \int_0^{+\infty} \left(\frac{x}{\theta^2} - \frac{1}{\theta} \right)^2 \frac{e^{-\frac{x}{\theta}}}{\theta} dx = \int_0^{+\infty} \frac{x^2}{\theta^4} \cdot \frac{e^{-\frac{x}{\theta}}}{\theta} dx -$$

$$- 2 \int_0^{+\infty} \frac{x}{\theta^3} \frac{e^{-\frac{x}{\theta}}}{\theta} dx + \int_0^{+\infty} \frac{1}{\theta^2} \frac{e^{-\frac{x}{\theta}}}{\theta} dx =$$

$$= \frac{2}{\theta^2} - \frac{2}{\theta^2} + \frac{1}{\theta^2} = \frac{1}{\theta^2} > 0 \quad \text{непр. } \forall \theta \in \Theta \subset \mathbb{R} \Rightarrow$$

$\Rightarrow$ можно регулярна

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2) а) $\tilde{\Theta}_1$ - меньше, тогда регулярна

$$2\tilde{\Theta}_1 = \frac{\Theta^2}{3} - \text{ср. на } \forall \text{ полноты, } \Theta \in \mathbb{R}$$

$\Rightarrow$ по гос. уст. оценка регулярна

б) $\tilde{\Theta}_2$ - меньше, тогда регулярна

$$2\tilde{\Theta}_2 = \frac{13}{25} \Theta^2 - \text{ср. на } \forall \text{ полноты, } \Theta \in \mathbb{R}$$

$\Rightarrow \tilde{\Theta}_2$ регулярна

$\Rightarrow$ неравенство $k-p$ работает где $\tilde{\Theta}_1, \tilde{\Theta}_2$

$$2\tilde{\Theta}_1 = \frac{\Theta^2}{3} = \frac{1}{n \cdot \frac{1}{\Theta^2}} = \frac{\Theta^2}{3} \Rightarrow \text{по гос. уст. оценка}$$

$$2\tilde{\Theta}_2 = \frac{13}{25} \Theta^2 > \frac{\Theta^2}{3} \Rightarrow \text{эсрессивная}$$

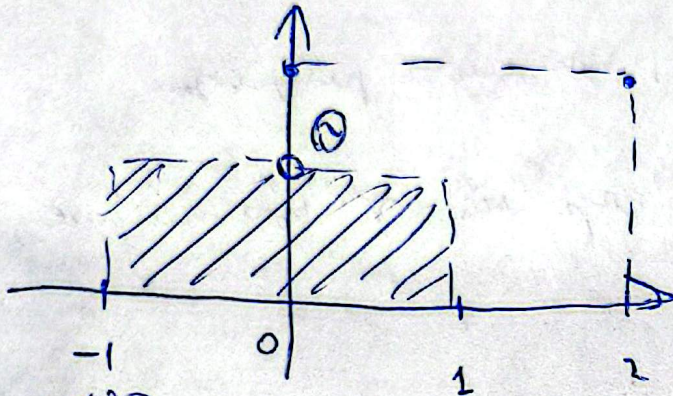
$\Rightarrow$ нельзя иметь сказать

но по $\textcircled{1}$ (! эср. оценка)

$\tilde{\Theta}_2$ не эсрессивная оценка.

Т.к.

$$f(x) = \vartheta \{(-1, 1) \setminus \{0\}\} + \left(\frac{1}{2} - \vartheta\right) \{0\} + \left(\frac{1}{2} - \vartheta\right) \{2\}, \quad \vartheta \in (0; \frac{1}{2})$$



$$a) M\zeta = \int_{-\infty}^{+\infty} x df(x, \vartheta) = \int_{-1}^1 \vartheta x dx + \left(\frac{1}{2} - \vartheta\right) \cdot 0 + \left(\frac{1}{2} - \vartheta\right) \cdot 2 = 1 - 2\vartheta$$

$$M\zeta^2 = \int_{-\infty}^{+\infty} x^2 df(x, \vartheta) = \int_{-1}^1 \vartheta x^2 dx + \left(\frac{1}{2} - \vartheta\right) \cdot 0 + \left(\frac{1}{2} - \vartheta\right) \cdot 4 = \frac{2}{3}\vartheta + 2 - 4\vartheta = 2 - \frac{10}{3}\vartheta$$

$$D\zeta = M\zeta^2 - M^2\zeta = 2 - \frac{10}{3}\vartheta - 1 + 4\vartheta - 4\vartheta^2 = 1 + \frac{2}{3}\vartheta - 4\vartheta^2; \quad \underline{QMM}$$

$$\alpha_1 = \alpha_1(\vartheta) = M\zeta, \quad 1 - 2\vartheta = \bar{x}$$

$$\tilde{\vartheta}_1 = \frac{1 - \bar{x}}{2}, \quad M\tilde{\vartheta}_1 = \vartheta: \quad M\tilde{\vartheta}_1 = M\left(\frac{1 - \bar{x}}{2}\right) = \vartheta =$$

ϑ_1 - несмещен.

$$D\tilde{\vartheta}_1 = D\left(\frac{1 - \bar{x}}{2}\right) = \frac{1}{4} D\bar{x} = \frac{1}{4} \frac{D\zeta}{n} = \frac{1 + \frac{2}{3}\vartheta - 4\vartheta^2}{4n} \xrightarrow{n \rightarrow \infty} 0 \approx 1 \quad \begin{matrix} \text{по} \\ \text{гос.} \\ \text{вл.} \end{matrix} : \vartheta_1 - \text{лост.}$$

$$\text{ОМН} \quad L(\Theta) = \prod_{i=1}^n p(x_i, \Theta) - \text{функция}$$

правдоподобия

и чисел $\{0; 2\}$, $n - m$ $(-1, 0) \cup (0, 1)$

$$L(\Theta) = \Theta^{n-m} \left(\frac{1}{2} - \Theta\right)^m$$

$$\ln L(\Theta) = (n-m) \ln \Theta + m \ln \left(\frac{1}{2} - \Theta\right)$$

$$(\ln L)' = \frac{n-m}{\Theta} + \frac{m}{\frac{1}{2} - \Theta}; \quad (n-m)\left(\frac{1}{2} - \Theta\right) - m\Theta = 0$$

$$(\ln L)'' = -\frac{(n-m)}{\Theta^2} - \frac{m}{\left(\frac{1}{2} - \Theta\right)^2} < 0 \Rightarrow \text{тогда макс}$$

$$\frac{1}{2} n - n\Theta = \frac{1}{2} m \Rightarrow$$

$$\tilde{\Theta}_2 = \frac{n-m}{2n} = \frac{1}{2} - \frac{D}{2}; \quad D = \frac{m}{n} \quad \text{ОМН}$$

б) ищ. на несмещ. и сост.

$$1) M\tilde{\Theta}_2 = M\left[\frac{1}{2} - \frac{1}{2}D\right] = \frac{1}{2} - \frac{1}{2}(1-2\Theta) = \Theta$$

$\Rightarrow \tilde{\Theta}_2$ - несмещ.

$$2) D\tilde{\Theta}_2 = \frac{1}{4}DD = \frac{p(1-p)}{4} = \frac{(1-2\Theta)2\Theta}{4n} \xrightarrow{n \rightarrow \infty} 0$$

$\Rightarrow$ дост. сост.

в) ищ. на эффективность

1) регулярность модели:

$p(x, \Theta)$ - непрерыв. дифер. по $\Theta \in (0; \frac{1}{2})$

$$\int_{-1}^1 \frac{\partial f(x, \theta)}{\partial \theta} + \sum_k p_k(\theta) = 0$$

$$\int_{-1}^1 dx + (-2) = 0 \quad \text{верно}$$

$$I(\theta) = M \left[\left(\frac{\partial \ln p(x)}{\partial \theta} \right)^2 \right] = \int_{-1}^1 \frac{1}{\theta^2} \theta dx + \left(\frac{1}{\frac{1}{2} - \theta} \right)^2$$

$$(1 - 2\theta) = \frac{2}{\theta} + \frac{4}{1 - 2\theta} = \frac{2}{\theta(1 - 2\theta)} \Rightarrow I(\theta) > 0$$

и $I(\theta)$ - вып. на $\theta \in (0, \frac{1}{2}) \Rightarrow$

можно регулярна

2) регулярность оценок

$\tilde{\theta}_1$ - вып., можно регул.

$$2\tilde{\theta}_1 = \frac{1 + \frac{2}{3}\theta + 4\theta^2}{4n} \quad \text{оц. на } \forall \text{ начеке}$$

оценка

регулярна

$\tilde{\theta}_2$ - вып., можно регулярна

$$2\tilde{\theta}_2 = \frac{(1 - 2\theta)2\theta}{4n} \quad \text{оц. на } \forall \text{ начеке}$$

оценка

регулярна

$\Rightarrow$ К-Р работает для $\tilde{\theta}_1$ и $\tilde{\theta}_2$

$$\tilde{\theta}_1: \frac{1 + \frac{2}{3}\theta - 4\theta^2}{4n} > \frac{\theta(1 - 2\theta)}{2n} \Rightarrow 1 + \frac{2}{3}\theta > 2\theta$$

$$\Rightarrow 1 - \frac{4}{3}\theta > 0, \theta \in (0; \frac{3}{4}) - \text{верно для}$$

$$\tilde{\theta}_2: \frac{(1 - 2\theta)2\theta}{4n} = \frac{\theta(1 - 2\theta)}{2n} \quad \text{по дост. уч.}$$

из 1

оц. эфр. оценка

$\tilde{\theta}_1$ - не является эфр.